THE BUILD / INKLOOM AI PROBLEM STATEMENT

Proofline

Evidence before approval.

A working review desk for brand content: connect the claim to its source, make the correction, and approve a version that cannot silently change.

HAIL MARY

ROUND 2 / WORKING BUILD

01

Review the claim

Text, PDF and image content enter the same review journey.

02

Keep the context

Findings cite source versions and offer approved wording.

03

Control the release

A different reviewer approves. Export rechecks validity.

THE DEMONSTRATION

A draft promises a guaranteed **30%** improvement. Its source supports **18%** on selected tasks. Correct it, approve it, then change the evidence: the previous approval is no longer valid.

Working product
https://proofline-hail-mary.dainty-shark-5340.chatgpt.site

Source repository
https://github.com/Prem-333/proofline-hail-mary

Bannari Amman Institute of Technology | Six participants
Round 2 build report | 24 October 2026

01 / PROBLEM AND USERS

Why approvals become outdated.

Content moves quickly. Its evidence, constraints and review history often do not.

The Inkloom AI brief asks for pre-publication brand governance across claims, language, visuals and accessibility, with explanations, corrections and human control. Proofline concentrates on one accountable journey: a creator prepares content; a reviewer checks the supporting material; release is tied to the reviewed version. [1]

Affected user	Failure addressed	What works now
Creator or agency partner	A number or promise drifts beyond the source.	Source-linked findings and optional corrections in the editor.
Brand owner	Terminology or approved facts change without reaching a campaign.	Versioned rules and evidence; earlier reviews become outdated.
Authorised reviewer	A file changes after sign-off.	Independent approval, fingerprints and an export-time validity check.
Audience	Promotions lose qualification or become hard to read.	Scope checks, image descriptions and selected colour contrast.

The distinctive decision

Approval records the reviewed content, supporting evidence and policy version. A change to any of these requires a fresh check. The source-change preview lets a brand owner inspect the expected draft findings before saving a revision.

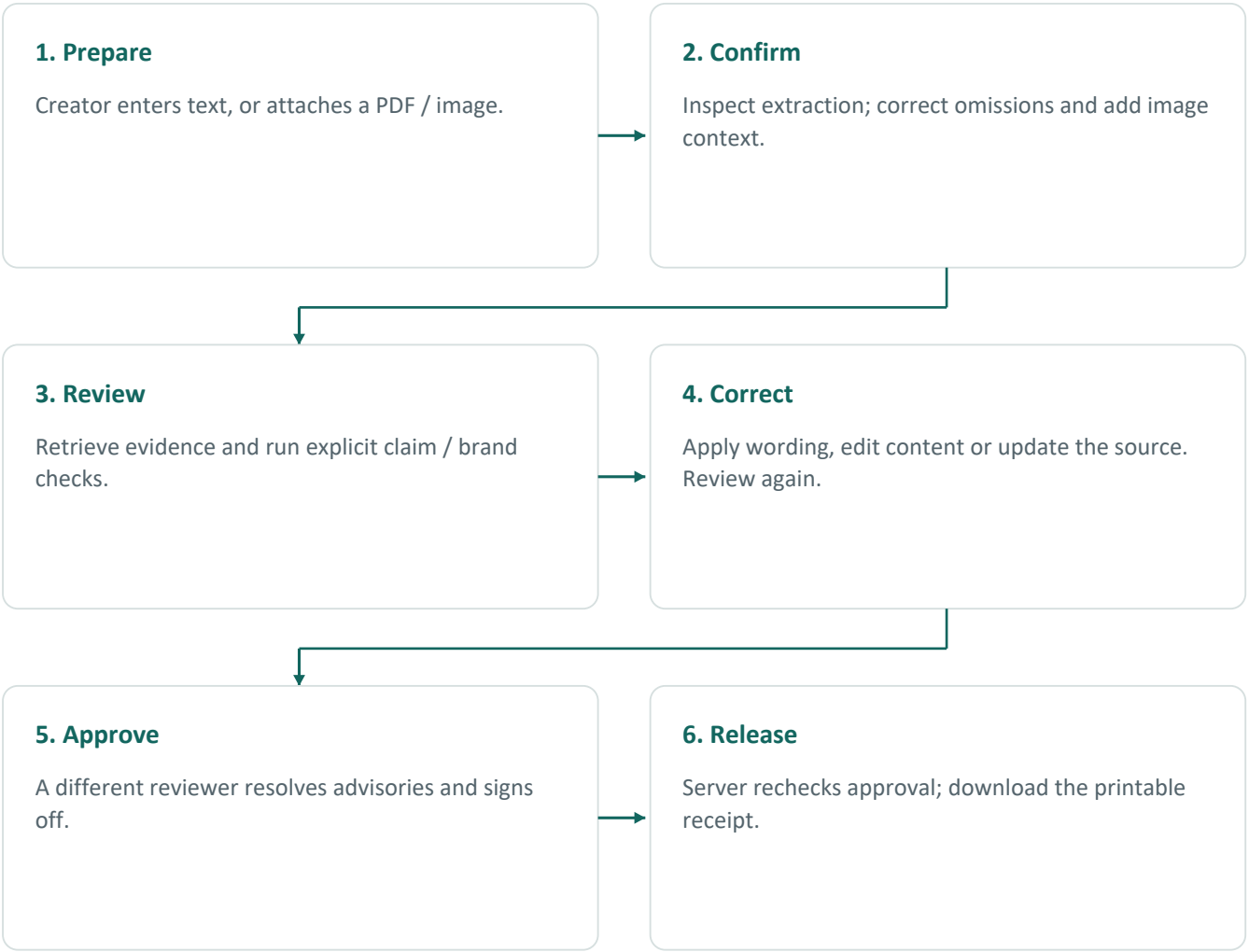
WHAT THE BUILD DOES NOT CLAIM

A matching source does not prove every sentence true. General factual accuracy, legal interpretation and full visual accessibility remain human review responsibilities.

02 / COMPLETE PRODUCT WORKFLOW

From draft to approved version.

The sample workspace can be used without an account or a model API key.



THE RELEASE GATE

Blocking findings cannot be waived. An advisory needs a correction or a reasoned reviewer exception. The creator cannot approve their own work, including when they are an administrator.

Invalidation loop: change content, a source or policy; run a new review; obtain fresh approval. The same checks run again when a receipt is exported.

03 / COMPLETED FEATURES

Implemented in Round 2.

A focused implementation of the core journey, with broader analysis reserved for Round 3.

Capability	Round 2 implementation	Boundary
Brand and evidence setup	Editable terminology, avoided terms, palette, contrast policy and scoped evidence.	Sources are manually defined; no automatic document-to-policy ingestion.
Three input formats	Text editor; digital PDF extraction; English image OCR; original file storage.	1,000 text words; PDF <=5 pages; PNG/JPEG <=10 MB.
Explainable findings	Numeric mismatch, scope, expiry, source disagreement, terminology and explicit tone terms.	General factual language can escape the detector.
AI-assisted retrieval	Browser MiniLM embeddings rank related excerpts; explicit checks follow.	Similarity is not entailment. Rules continue if the model fails to load.
Corrections and collaboration	Apply approved wording, comment, request changes and inspect previous revisions.	No automatic rewriting of original PDF/image files.
Human approval	Different approver; reasoned advisory exceptions; non-waivable blockers.	Demo personas are simulated. Private membership uses signed-in identities.
Version-bound release	Content hash, source digest, policy version and fresh server checks at export.	Receipt is point-in-time evidence, not a digitally signed certificate.
Operational visibility	Live draft counts and an activity record of review decisions.	Recurring-rule analytics are a Round 3 improvement.

The brief supports multiple formats and human control. This build delivers those foundations without presenting the full final-round scope as complete.

04 / SOURCE-CHANGE PREVIEW

See the effect before saving.

A brand owner can compare draft findings against a proposed source revision.

Saved source / v1

The internal study supports 18% less reporting time on selected tasks.

→

Proposed source / v2

A revised study supports 12% less reporting time on the same tasks.

DRAFT UNDER REVIEW

“Our internal study measured 18% less reporting time on selected tasks.”

Preview result	What the owner sees
Blocking findings: 0 to 1	The draft still says 18%. The proposed source supports 12%.
Before / after findings	New and resolved findings, quoted text and the rule explanation for each draft.
No saved changes	Previewing does not edit the source, drafts, review history or approval records.
Save remains a separate action	Saving the revision makes existing reviews outdated. Content must be checked and approved again.

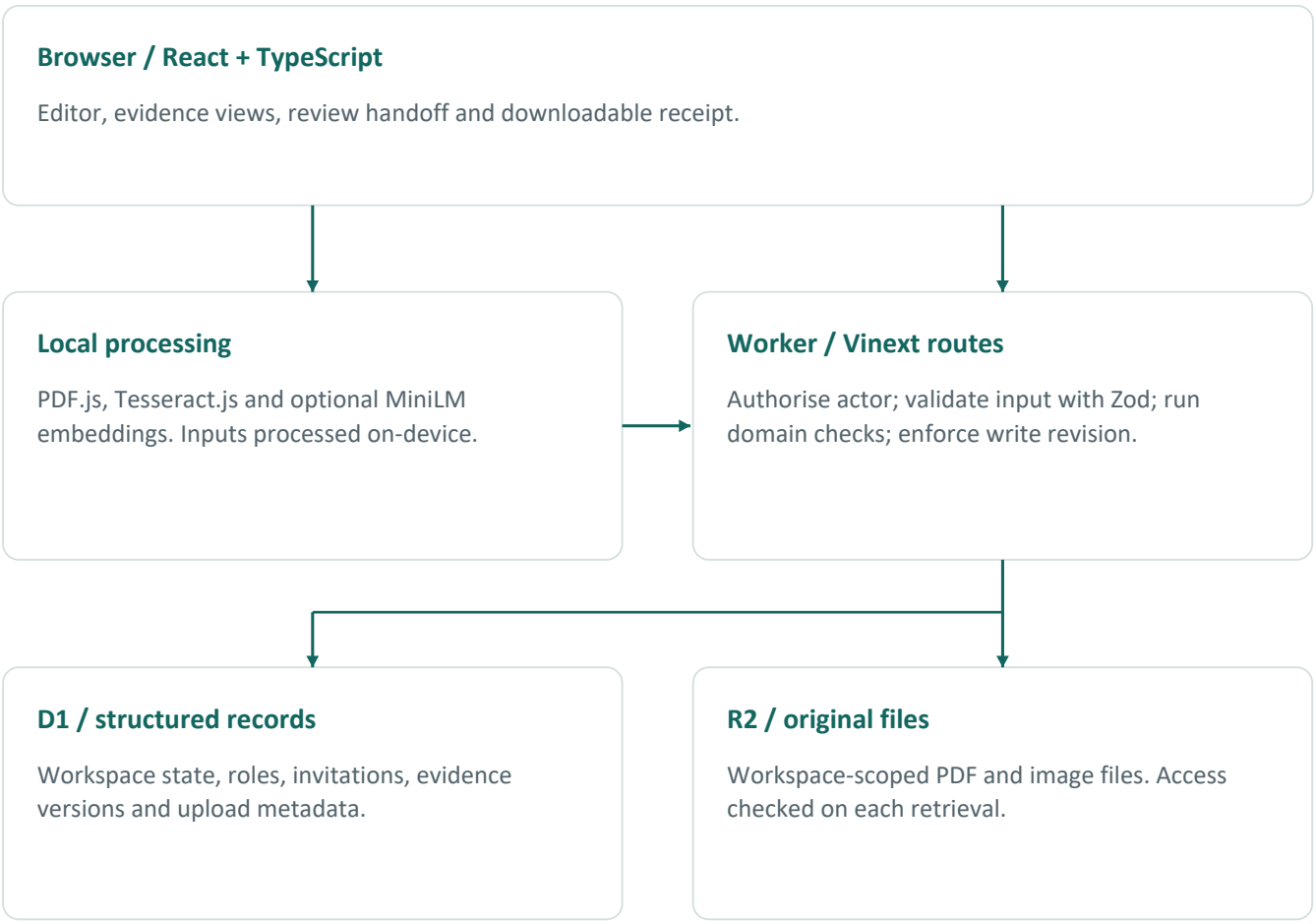
The preview runs the same explicit rules against both source versions. It uses saved drafts and lexical matching; it does not run the optional embedding model or grant approval.

Round 3 boundary: this is an on-demand comparison. Scheduled freshness checks, dependency indexing and targeted re-review queues remain planned work.

05 / TECHNICAL DEPTH

Architecture and release checks.

The hardest part is preserving approval validity while content, rules and sources change.



How the release gate stays consistent

1. A review stores a SHA-256 content fingerprint, policy version and source digest.
2. Approval checks the current values, resolved findings and a different reviewer.
3. Export repeats those checks. Concurrent writes use a revision comparison and return HTTP 409 instead of silently overwriting.

Trade-off: the current digest covers all workspace evidence, so an unrelated source revision can also require re-review. Round 3 replaces this conservative approach with claim-to-source dependencies.

06 / VERIFICATION EVIDENCE

Verification results.

Observed implementation results. These are not detection-accuracy or user-study scores.

AUTOMATED SCENARIOS PASSED

Domain behaviour



29 / 29

HTTP journey and access



11 / 11

Bar length shows scenario count on a shared 0-29 scale. Both groups passed in full.

40 distinct automated scenarios: 29 domain tests and 11 HTTP checks. The HTTP test runner also reports a parent suite; it is not counted as an additional scenario.

Verification	Observed result
Approval and export	Create, analyse, submit, independently approve and export completed through the API and browser.
Invalidation and isolation	Content/source/policy changes revoke validity; self-approval, cross-session reads and stale writes are rejected.
Real browser integrations	Local MiniLM review returned evidence matches; PDF text extracted; image OCR returned the fixture text with 95% engine confidence.
Input and extraction gates	Unsupported file signature rejected; unconfirmed extraction blocked approval; confirmed image with description passed explicit checks.
Source-change preview	Five added scenarios cover new blockers, resolved findings, expiry, unrelated sources and extracted attachment claims. Existing hosted HTTP checks: 11 passed on 23 October.

OCR confidence is an engine output for one synthetic image, not measured product accuracy. The Round 1 held-out corpus, real-user sessions and performance measurements still need to be completed.

07 / CHANGES TO THE ORIGINAL PLAN

Implementation decisions.

The implementation reduces deployment and API-key dependencies for a usable judging experience.

Round 1 plan	Round 2 choice	Reason / consequence
Next.js + FastAPI	React / TypeScript, Vinext and Worker routes	One deployable app and shared domain checks.
PostgreSQL / Supabase	Cloudflare D1 + R2	Hosted records and original-file storage within the deployment.
Gemini structured generation	Local MiniLM retrieval + explicit rules	No model API key; corrections use approved wording. Broader natural-language judgement remains limited.
Server pypdf / Tesseract / Pillow	Browser PDF.js + Tesseract.js	No separate processing service; users confirm extracted text.
Broad visual compliance	Description, palette and entered colour-pair checks	Full image-region analysis is deferred and clearly labelled.
Designed report package	Printable HTML approval receipt	Preserves fingerprints, findings, evidence and sign-off; can be saved as PDF.

Privacy and operational limits

Private workspaces enforce signed-in membership. Demo workspaces use an isolated session cookie and expire after seven days. Uploads are checked and scoped to the workspace; request and storage limits constrain usage. Raw documents are not placed in routine application logs.

Text, extracted content and originals are stored by the application. OCR and embeddings run in the browser; external hosts supply their runtime/model files. A confidential pilot still needs a retention/deletion policy, provider review and security assessment.

08 / ROUND 3 COMMITMENTS

Round 3 development plan.

Each extends a working Round 2 baseline and has a demonstrable acceptance condition.

01 Evidence freshness and affected content

Today: manual source-change preview and on-demand validity checks.

Build a claim-to-source dependency index, scheduled expiry checks and an affected-drafts queue.

Acceptance: Changing or expiring a source identifies every dependent approved draft. The dependency index leaves unrelated drafts unflagged.

DELIVERY OWNERS / Kanish + Premnath

02 Visual evidence overlays

Today: plain OCR text and creator-entered colour values.

Retain word boxes and confidence; locate findings on the image and route uncertain text into a correction queue.

Acceptance: A reviewer can locate and correct each flagged phrase. Low-confidence regions require explicit review before release.

DELIVERY OWNERS / Bharathraj + Mirthika

03 Rule clarification from review decisions

Today: comments and individual reasoned exceptions.

Group repeated exceptions by rule, show examples and let the brand owner test and approve a policy revision.

Acceptance: Repeated exceptions create a traceable clarification item. No exception or model suggestion changes policy automatically.

DELIVERY OWNERS / Rithikha + dharshini

09 / TEAM AND DELIVERY OWNERSHIP

Hail Mary

Bannari Amman Institute of Technology | Third year | Computer Science and Business Systems

Participant	Assigned ownership	Required verification
Premnath V R Team leader	Product scope, integration, demo journey and release decisions.	Confirm integration review and the end-to-end demonstration.
Mirthika S	Frontend, usability and interface accessibility.	Review desktop/mobile flow, forms and keyboard behaviour.
Rithikha B	Evidence retrieval, model evaluation and test corpus.	Review claim cases and document retrieval errors.
Kanish S	Backend, authorization, persistence and security.	Review access checks, concurrency and approval invalidation.
Bharathraj Nagarajan	PDF/image extraction and visual analysis.	Review extraction failures and source-image comparisons.
dharshini T	Quality assurance, analytics and release verification.	Record regression results, link checks and submission readiness.

CONTRIBUTION STATUS

These responsibilities carry forward the Round 1 allocation. Individual completed contributions have not yet been verified by the team. The initial implementation and documentation were prepared with Codex assistance; the repository includes a contribution log for accurate member sign-off.

Event window supplied by the team: 20-30 October 2026, completely online. Round 1 established the plan; Round 2 delivers the core workflow; Round 3 delivers the three additions and reliability evidence. Specific round cutoffs follow organiser announcements.

10 / SUBMISSION, IMPACT AND NEXT STEPS

Project links and next steps.

Start with the sample journey, then inspect the code and release safeguards.

Working project
https://proofline-hail-mary.dainty-shark-5340.chatgpt.site

Source repository
https://github.com/Prem-333/proofline-hail-mary

Expected impact and future scope

The intended benefit is fewer avoidable review cycles and fewer releases supported by stale or mis-scoped evidence. This has not yet been measured with real teams. After the event, pilot with a small brand/agency team, measure review time and false alarms, then consider multilingual support, document connectors, stronger asset checks and enterprise retention controls.

Submission integrity

Codex assisted architecture, code, tests, writing and layout. Reused components include the Sites Vinext starter, React, Cloudflare tooling, Drizzle, Zod, Lucide, PDF.js, Tesseract.js, Transformers.js and the pretrained MiniLM model. Exact package versions are in the lockfile; attribution and limitations are in the repository. Synthetic fixtures are labelled. No user interviews, time savings or detection accuracy are invented.

Source notes

[1] Inkloom AI brief - Problem_statement.pdf, pp. 1-5; Round 2 organiser message supplied by the team.

[2] MiniLM model card - ONNX embeddings used for local semantic retrieval. Apache-2.0 model licence.

[3] PDF.js and Tesseract.js - Document extraction and browser OCR; third-party libraries.

[4] Tesseract.js upstream - English OCR runtime and language data.

[5] W3C contrast guidance - Basis for the supplied colour-pair calculation; not a full accessibility audit.

Reviewed 24 October 2026. Before form submission: verify each participant contribution, upload this PDF to Drive with Viewer access for anyone with the link, and test the final Drive link in an incognito window.